COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	9/23/2026	**DATE SUBMITTED:**	9/23/2026
NAME:	Rameses Andrei James Domingo	**INSTRUCTOR:**	ENGR. Robin Valenzuela

ACTIVITY NO. ___ **Midterm Skills Exam**

1. PROCEDURE OUTPUT

MIDTERM_DOMINGO.cpp > main()

```cpp
#include <iostream>
#include <string>
#include<cctype>
using namespace std;

int main () {
    int N;

    cout << "How many placements would you like to do (1 - 32): ";
    cin >> N;

    if (N >= 1 && N <= 32) {
        string board[8][8];

        for (int g = 0; g < 8; g++ ) {
            for (int w = 0; w < 8; w++) {
                board[g][w] = ".";
            }
        }
        int r, c, t = 0, h = 0;
        char p;


        for (int y = 0; y < N; y++) {
            cout << "row column piece (R, N, B, Q, K, P, r, n, b, q, k, p): ";
            cin >> r;
            cin >> c;
            cin >> p;

            board[r][c] = p;

            if (isalpha(p)) {
                if (isupper(p)) {
                    t++;

                }else if (islower(p)) {
                    h++;

                }

            }

        }
```

```
        for (int n = 0; n < 8; n++ ) {
            for (int e = 0; e < 8; e++) {
                cout << board[n][e] << "\t";
            }
            cout << endl;
        }
        cout << "Whitepieces: " << t << "\t" << "Blackpieces: " << h << endl;

        for (int a = 0; a < 8; a++ ) {
            int c = 0;

            for (int b = 0; b < 8; b++) {
                if (board[a][b] != ".") {
                    c++;

                } else {}
            }

            cout << "Row " << a << ": " << c << endl;

        }

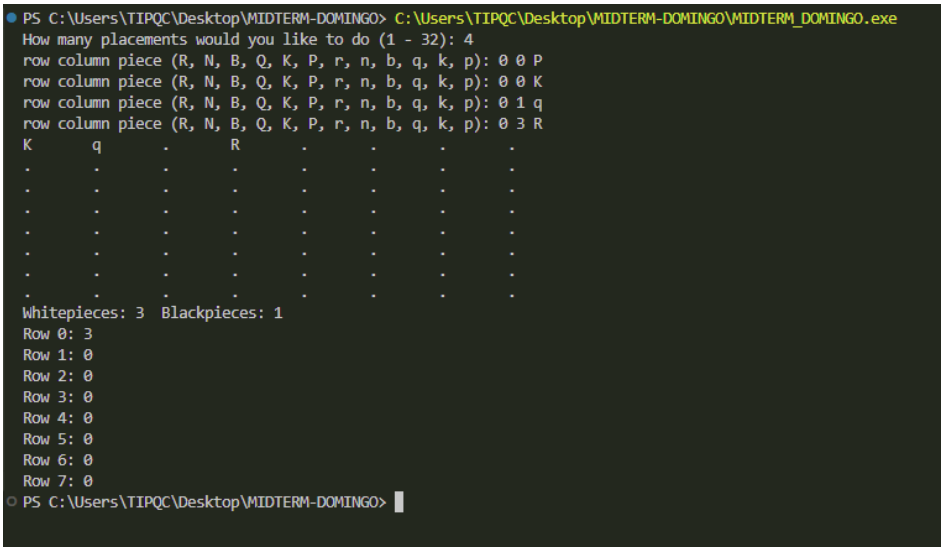

    } else {
        cout << "INVALID NUMBER, 1 - 32 ONLY!" << endl;
    }



    return 0;
}
```

EXPLANATION: This program prompts the user to input an amount of letters that he/she will input in a chess board wherein each input gets allocated in the pre-made 2d array for the chess board that changes values based on the position and equivalent letter that the user wants to input using a row, column, piece pattern.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

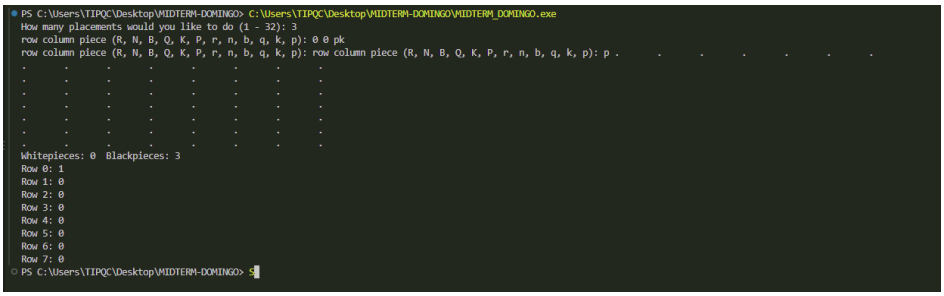
Case	Description
Case 1	Sample input with an overwritten square
In this snippet we can see that the user placed letters in the same position but the program still counted it as one square.	

```
PS C:\Users\TIPQC\Desktop\MIDTERM-DOMINGO> C:\Users\TIPQC\Desktop\MIDTERM-DOMINGO\MIDTERM_DOMINGO.exe
How many placements would you like to do (1 - 32): 4
row column piece (R, N, B, Q, K, P, r, n, b, q, k, p): 0 0 P
row column piece (R, N, B, Q, K, P, r, n, b, q, k, p): 0 0 K
row column piece (R, N, B, Q, K, P, r, n, b, q, k, p): 0 1 q
row column piece (R, N, B, Q, K, P, r, n, b, q, k, p): 0 3 R
K       q       .       R       .       .       .       .
.       .       .       .       .       .       .       .
.       .       .       .       .       .       .       .
.       .       .       .       .       .       .       .
.       .       .       .       .       .       .       .
.       .       .       .       .       .       .       .
.       .       .       .       .       .       .       .
.       .       .       .       .       .       .       .
Whitepieces: 3  Blackpieces: 1
Row 0: 3
Row 1: 0
Row 2: 0
Row 3: 0
Row 4: 0
Row 5: 0
Row 6: 0
Row 7: 0
PS C:\Users\TIPQC\Desktop\MIDTERM-DOMINGO>
```

Case 2

Multiple pieces placed in the same row

Since the program is using char function, the output resulted in an error since it detected multiple letters

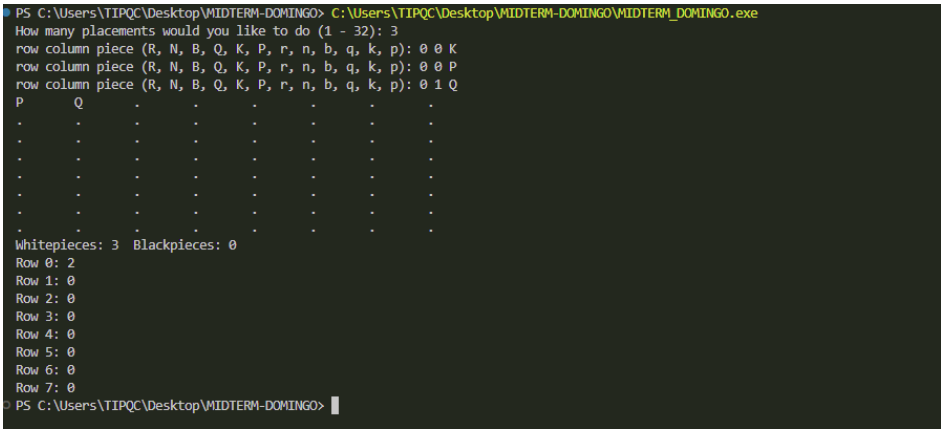


Case 3

Multiple rows tied for the highest number of pieces

Case 4

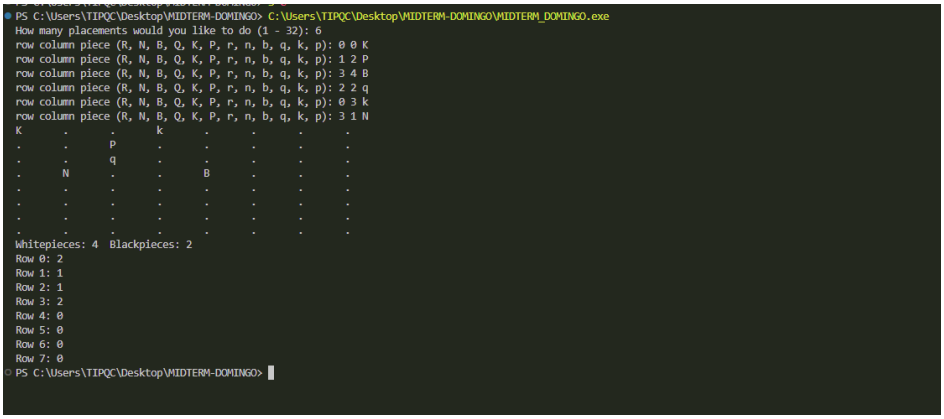
Placements that overwrite both White and Black pieces



Case 5

A board with pieces distributed across different rows

This is the program output with the scattered pieces.



Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.